Personalized Learning AI

An Adaptive Deep Learning-Based Student Learning & Recommendation System

Project Domain: Deep Learning, Adaptive Learning, Knowledge Tracing and Intelligent Tutoring Systems

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First Review Sections

- Abstract
- Introduction
- Literature Survey - Methods Used
- Flowchart & System Architecture
- Deep Learning Models

Project goal: Build an AI learning system that understands each student's progress and changes the learning path according to the student's needs.

2. ABSTRACT

Point-wise summary of the proposed system.

1. Problem

- Traditional e-learning platforms often provide the same sequence of lessons and questions to every student.
- Students have different learning speeds, strengths, weaknesses and revision needs.

2. Proposed Solution

- The system acts like an AI teacher that continuously observes a student's learning behavior.
- It estimates what the student currently understands in each topic.

3. Information Collected

- Question/topic, correctness, time taken, attempts, hints and previous performance.

4. Deep Learning

- An LSTM or GRU sequence model learns patterns from the student's series of question attempts.
- The model produces a topic-wise knowledge profile.

5. Personalization

- Weak topics are identified and strong topics can move to more challenging questions.
- Question difficulty is adjusted according to performance.

6. Revision / Forgetting Support

- The system compares current and older performance and can recommend revision when an earlier topic appears to be forgotten.

7. Final Output

- The student receives a personalized learning path, suitable questions and revision suggestions.

In one line: Student solves questions → AI studies the learning behavior → AI finds weak areas → AI recommends what the student should learn next.

3. INTRODUCTION

Students do not learn at the same speed. A fixed course can therefore be useful for one student but frustrating for another.

Problems with the normal approach

- A student who already understands a topic may receive too many basic questions.
- A student who is struggling may be pushed to a harder topic too early.
- Final marks do not show the complete learning process.
- Two students with the same score can have very different weaknesses.
- A previously learned topic may be forgotten after a long gap.

What our system tries to understand

- What topics does the student know well?
- Which topics are weak?
- Is the student answering correctly but taking too much time?
- Does the student need easier practice or a harder challenge?
- Which older topics need revision?

Example

Student A: Arrays 90%, Strings 85%, Recursion 42%. The system recommends recursion revision instead of more advanced questions.

Student B: Arrays 55%, Recursion 82%. The system gives more array practice, even though Student A does not need it.

Main objective

To create an adaptive learning platform that uses a student's interaction history to estimate knowledge and provide a learning path that changes with the student.

4. LITERATURE SURVEY - METHODS USED

The literature survey is arranged from classical knowledge tracing to Deep Learning and recent process-aware models. Each method and past work is summarized with a direct research link.

Method	How researchers did it	Limitation	Research link
IRT Item Response Theory	Relates student ability to question difficulty and estimates the chance of a correct answer.	Student ability is treated as relatively static.	Open IRT source
BKT Bayesian Knowledge Tracing	Tracks whether a student has mastered a concept from a sequence of correct and incorrect answers.	Usually models concepts separately and uses limited behavior information.	Open BKT paper
DKT Deep Knowledge Tracing	Uses RNN/LSTM-style Deep Learning to learn patterns from a sequence of student-question interactions.	Basic versions mainly focus on question/answer sequences.	Open DKT paper
DKVMN Dynamic Key-Value Memory Network	Uses a memory representation of concepts and updates the student's knowledge state after interactions.	More complex for a small student implementation.	Open DKVMN paper

The following past works provide the research background for the ideas used in our project.

Year / Paper	What they did	Limitation / relevance	Link
1995 Corbett & Anderson Knowledge Tracing: Modeling the Acquisition of Procedural Knowledge	Introduced a probabilistic knowledge-tracing approach for estimating a student's changing knowledge while learning.	Classical approach; motivates our knowledge-tracing foundation.	Open paper
2009 Pavlik, Cen & Koedinger Performance Factors Analysis	Used practice-history factors to model learning and support adaptive practice.	Not a Deep Learning sequence model, but useful for adaptation from practice history.	Open paper
2010 Pardos & Heffernan Modeling Individualization in a Bayesian Networks Implementation of Knowledge Tracing	Introduced student-specific prior information and individualized knowledge estimation.	Bayesian approach; supports our idea of treating students individually.	Open paper
2011 Pardos & Heffernan KT-IDEM: Introducing Item Difficulty to the Knowledge Tracing Model	Extended knowledge tracing by explicitly considering item difficulty when predicting performance.	Shows that question difficulty can be included in knowledge tracing.	Open paper

Why these papers matter to us: They establish the ideas of tracking changing knowledge, individualizing the student model, and considering question difficulty.

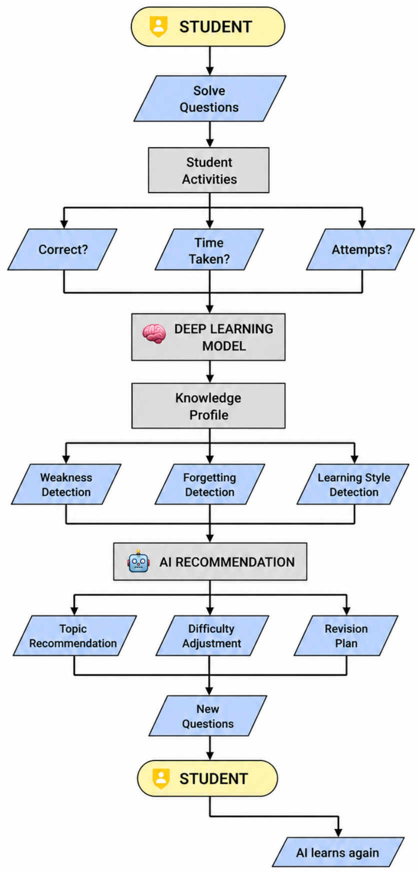
What we learn from the literature

- Student knowledge should be updated as the student answers more questions.
- Sequence-based Deep Learning is useful because learning changes over time.
- Simple behavior signals such as time and attempts can provide more information than final marks alone.
- Our project uses these ideas in a simpler, practical implementation.

Literature-to-project link: We use the main idea of Deep Knowledge Tracing, then add simple student behavior signals and a recommendation layer.

5. FLOWCHART & SYSTEM ARCHITECTURE

The flow is based on the second reference image supplied by the team. It shows the feedback loop between the student, the AI model and personalized questions.



Step	What happens?
1. Student	Starts learning and solves questions.
2. Activities	System records correctness, time and attempts.
3. Deep Learning	Model learns from the sequence of interactions.
4. Knowledge Profile	System estimates topic-wise understanding.
5. Analysis	Weakness, forgetting and learning patterns are identified.
6. Recommendation	AI chooses topic, difficulty and revision.
7. New Questions	Student receives personalized questions.
8. Feedback Loop	New result is stored and the profile is updated.

6. DEEP LEARNING MODELS

For the first review, the model is explained without mathematical equations or complicated implementation details.

Model	Simple meaning	Role in our project
RNN	Processes information in a sequence and uses earlier steps while handling the next step.	Basic idea for processing a student's sequence of attempts.
LSTM	A recurrent model that can keep useful information across longer sequences.	Can estimate how student knowledge changes across many questions.
GRU	A simpler recurrent model that learns which previous information to keep or update.	Practical alternative to LSTM for knowledge tracing.

Recommended model

LSTM / GRU-based Knowledge Tracing
Input: sequence of student question interactions.
Processing: model learns patterns in the sequence.
Output: estimated knowledge level for different topics.
Use: recommendation module decides what the student should do next.

Why Deep Learning?

- Student learning is a sequence: question after question, performance changes.
- The model can use previous interactions instead of looking at only the latest score.
- It can produce a changing knowledge profile as new answers are collected.

Simple example

Q1: Recursion - correct → Q2: Recursion - wrong → Q3: Recursion - wrong → Q4: Recursion - correct.
The model uses this sequence to update the estimated knowledge instead of treating each question independently.

Thank you